



Abstraction and Knowledge in Reinforcement Learning: Evaluating Concept Grounding on Craftax

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Introduction

Proximal Policy Optimization (PPO) [1] learns slowly on the challenging Craftax benchmark [2], where one source of difficulty is a wide variety of grid tiles that are discovered progressively as the agent's gameplay improves. As neuro-symbolic architectures can improve reinforcement learning [3], we investigated learning with expert guidance in Craftax. We found that agents able to group related tiles completed more achievements in a rare level, and that guidance with prior knowledge reduced sensitivity to initial conditions.

Hypothesis 1 (H1)

Learning from abstracted inputs increases generalization.

We tested our hypotheses using our concept grounding method: by forcing agents to generalize across tiles using a compression bottleneck, and by using an auxiliary multi-label classification loss to ground tile embeddings in an expert-defined taxonomy of Craftax concepts.

Methods

Concept grounding is an encoder-decoder method for gridworlds with symbolic tiles. In Craftax, the encoder maps 83D tiles to 32D embeddings in an observation preprocessing step. During training, the decoder maps 32D embeddings to 19 concept probabilities (sigmoid activation), the ground truth is computed with expert-provided binary propositions, and a standard binary cross-entropy loss for imbalanced multi-label classification is added to the standard PPO loss.

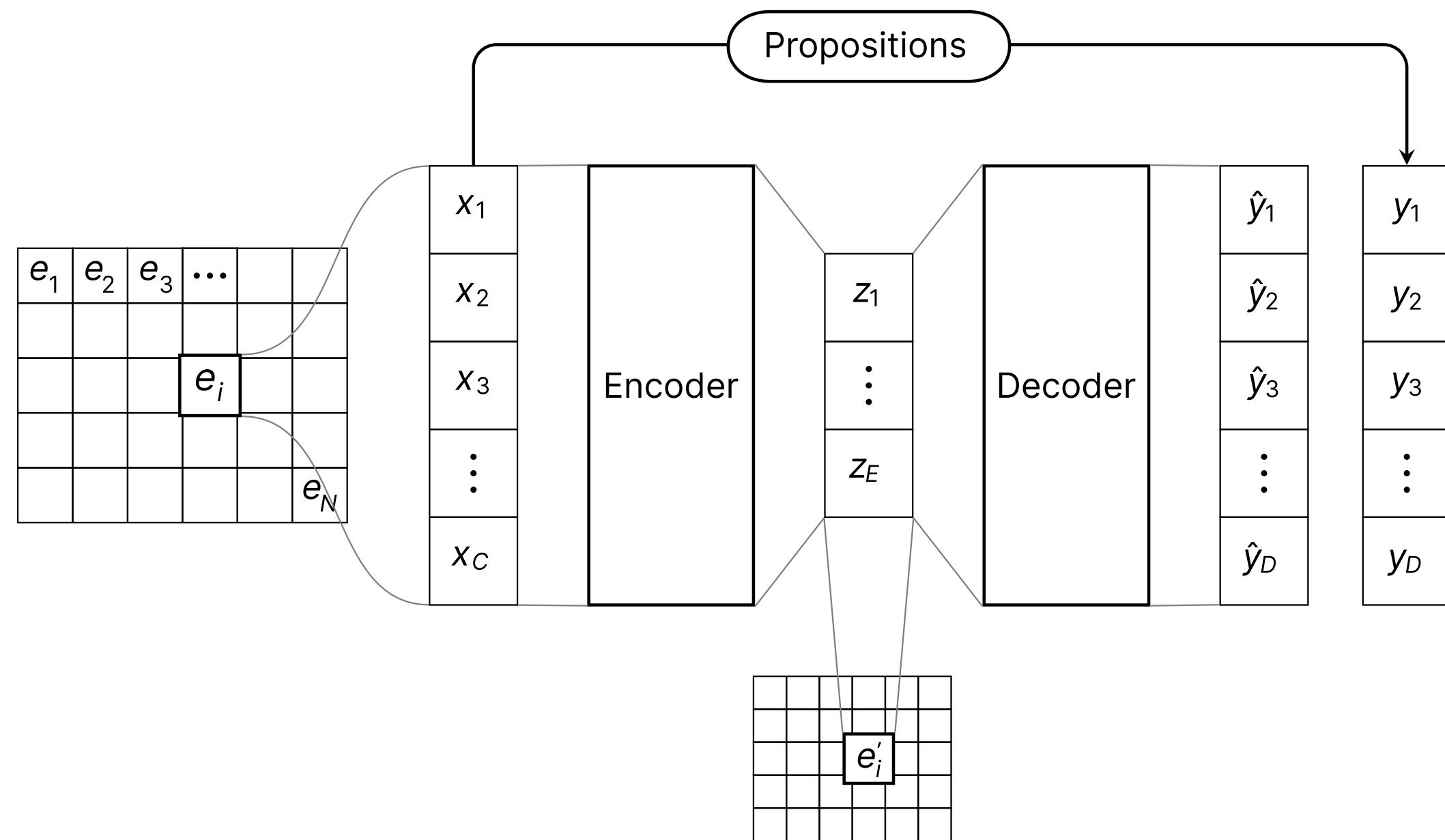


Figure 1: Concept grounding architecture for gridworlds

We designed a taxonomy of 19 Craftax tile concepts and implemented it with efficient tensor operations. Each leaf is a tile dimension compared to 1 unless noted, and each concept is a disjunction of other concepts or leaves.

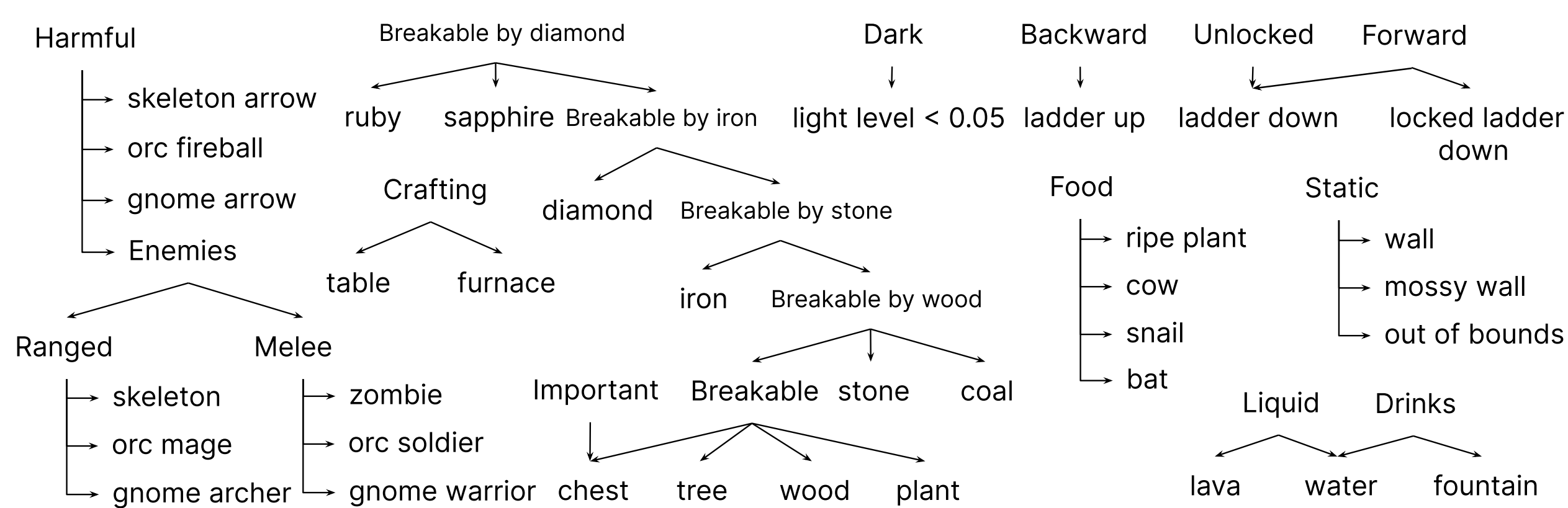


Figure 2: Proposed Craftax taxonomy

We evaluated the effect of abstraction with compression (**H1**), the effect of alternative guidance by instead learning to reconstruct tiles (**H2**), and the effect of bad priors by using a randomized taxonomy.

For each method, we trained 20 agents for 500 million steps, starting from different random seeds, and evaluated each one for 100 thousand episodes. During evaluation agents sampled their policy without using the decoder. We reported the average and maximum episode score (mean and standard deviation across agents), and the selected achievement completion rates (mean, 95ci, and all values). In the paper we tested for differences in completion rates of 42 achievements using a two-sided U test with Bonferroni-Holm correction.

Baseline

Non-recurrent PPO agent learns from raw observations.

Compression

The encoder compresses tiles, but the decoder is removed.

Autoencoder

Compression, but a symmetric decoder reconstructs tiles.

Grounding

Compression, but a single-layer decoder classifies concepts.

Shuffled

Grounding, but the leaves of the taxonomy are randomly shuffled.

Results

Embedding tiles halved the total agent parameter count without decreasing the average score. It also increased the maximum score and training times.

Table 1: Evaluation summary

Method	Tiles	Params	Avg. Score	Max. Score	Train Time
Baseline	83D	9.54M	11.62%±0.13%	22.75%±0.41%	4.6h±0.6h
Compression	32D	4.39M	11.82%±0.19%	25.24%±1.00%	6.7h±0.2h
Autoencoder	32D	4.42M	11.77%±0.17%	25.27%±1.20%	7.9h±0.9h
Shuffled	32D	4.39M	11.65%±0.16%	25.31%±1.90%	6.1h±0.5h
Grounding	32D	4.39M	11.87%±0.18%	26.42%±1.15%	6.2h±0.4h

Emeddings significantly increased achievement rates for analogous eat and defeat achievements ($\alpha < 0.05$), but caused random failures, particularly in make iron achievements, that we partially mitigated with grounding.

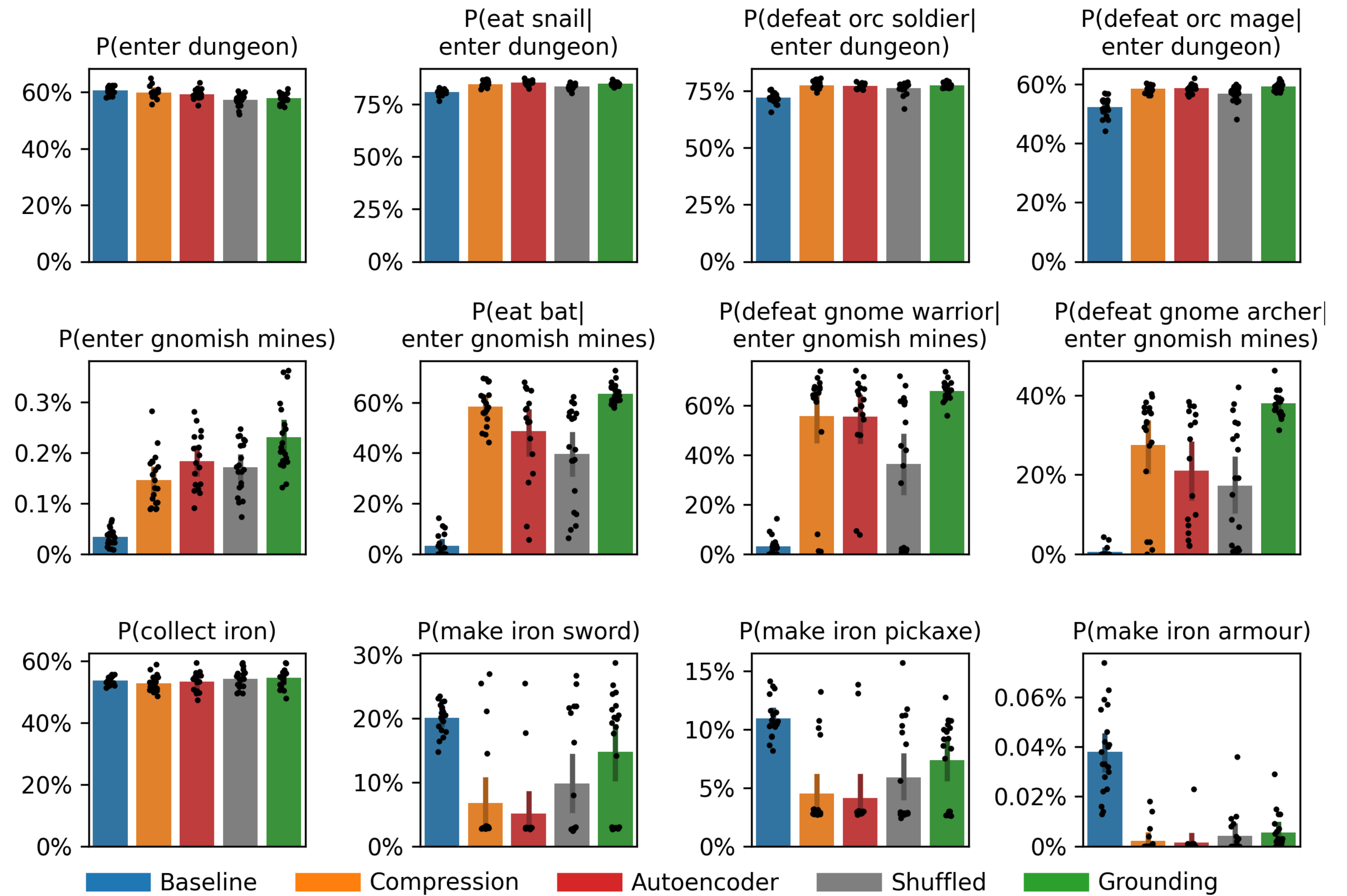


Figure 3: Selected achievement completion rates

We visualized the global embedding structure with UMAP, but failure-case analysis did not clearly indicate the root cause of instabilities.

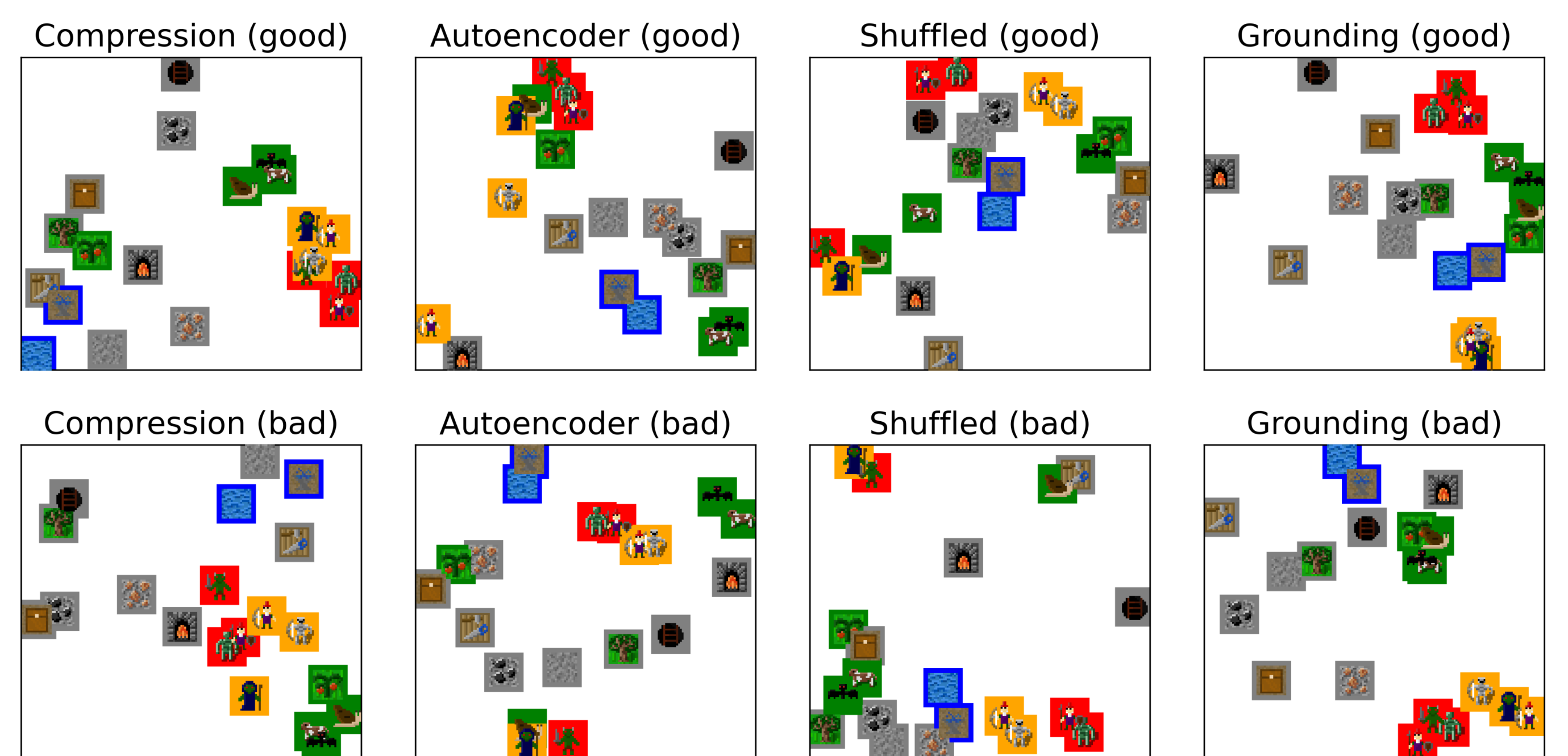


Figure 4: Tile embedding UMAP (n_neighbors = max)

Conclusions

Embedding agents abstracted over tiles, significantly increasing analogous achievement rates, with dramatic improvements in rare scenarios (**H1**). Jointly learning multi-label classification stabilized achievement rates in rare scenarios and recovered some performance lost in a bottleneck (**H2**). Supervised reconstruction failed to stabilize learning in a symbolic gridworld where tiles lack visual patterns. Misleading guidance mainly hurt rare-level performance without preventing learning. **Limitations:** concept grounding assumes a symbolic environment and requires knowledge engineering. **Future work:** recurrent architectures, richer knowledge bases, automated knowledge engineering, knowledge transfer. **Special thanks** to Jędrzej Potoniec and Wojciech Kotłowski for advice and feedback, and to the Poznań University of Technology for GPU access and financial support (SBAD/0770).

[1]: John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms, 2017. URL <https://arxiv.org/abs/1707.06347>.

[2]: Michael T. Matthews, Michael Beukman, Benjamin Ellis, Mikayel Samvelyan, Matthew Thomas Jackson, Samuel Coward, and Jakob Nicolaus Foerster. Craftax: A lightning-fast benchmark for open-ended reinforcement learning. In ICML, Proceedings of Machine Learning Research, pages 35104–35137. PMLR, 2024.

[3]: Kamal Acharya, Waleed Raza, Carlos M. J. M. Dourado Júnior, Alvaro Velasquez, and Houbing Herbert Song. Neurosymbolic reinforcement learning and planning: A survey. IEEE Transactions on Artificial Intelligence, 5(5):1939–1953, 2024.